

Titanic Survival Prediction Using Ensemble Boosting Models

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Introduction

This assignment focuses on predicting passenger survival on the Titanic using machine learning ensemble models. The Titanic dataset contains passenger information such as class, gender, age, number of siblings, number of parents, and ticket fare. The objective is to train multiple boosting models and compare their performance to determine which model predicts survival most accurately.

Models Used

Three tree-based boosting models were trained:

- XGBoost
- LightGBM
- CatBoost

All models were trained using the following parameters:

- `n_estimators = 100`
- `learning_rate = 0.1`
- `max_depth = 5`

Model Performance

Model	Accuracy
XGBoost	0.810
LightGBM	0.821
CatBoost	0.793

Among the tested models, LightGBM achieved the highest accuracy of 82.1%, making it the best performing model for this task.

Confusion Matrix Interpretation

The confusion matrix of the best model shows:

- 92 passengers correctly predicted as not survived

- 55 passengers correctly predicted as survived
- 13 passengers incorrectly predicted as survived
- 19 passengers incorrectly predicted as not survived

This indicates that the model performs well in identifying both survival and non-survival cases.

What I Learned

Through this assignment, several important machine learning concepts were learned:

- Data loading and exploration using Pandas
- Handling missing values using median replacement
- Encoding categorical variables into numeric form
- Splitting datasets into training and testing sets
- Training ensemble boosting models
- Evaluating models using accuracy, precision, recall, and F1-score
- Visualizing results using confusion matrices

Challenges Faced

During the implementation process, several challenges were encountered:

- Installing LightGBM due to dependency issues
- Handling missing values in the *Age* and *Fare* features
- Converting categorical variables such as gender into numeric values

These issues were resolved through proper preprocessing and package installation.

Conclusion

The results show that ensemble boosting algorithms are effective for tabular datasets. Among the models tested, LightGBM achieved the best accuracy. Its efficient gradient boosting framework and leaf-wise tree growth strategy allow it to learn complex relationships in the dataset effectively.